Fiscal Year:	FY 2010	**Task Last Updated:**	FY 08/20/2009
PI Name:	Oziomek, Thomas B.S.		
Project Title:	Total System Approach		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2008	**End Date:**	10/01/2009
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2009, as project ended early. Per M. Perchonok/JSC (12/13/2010)		
Key Personnel Changes/Previous PI:	NOTE PI changed to Thomas Oziomek shortly after project started (information from M. Perchonok/JSC 8/2009).		
COI Name (Institution):	Perchonok, Michele (NASA Johnson Space Center)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	NASA has established the goal of returning human expeditions to the moon and extending exploration to Mars. Extended manned missions of these types require massive quantities of food to be flown into space. This presents two challenges when dealing with a food system. These challenges are maintaining the quality of the food throughout its shelf life which may be in excess of five years, and to assure the mass and volume of the food system are minimized. A Total System Approach is an operating system that considers each parameter of an overall process and seeks to optimize each of these parameters to maximize the output. In the case of the NASA food system, the output would be a food system with low mass packaging using high barrier packaging materials. The purpose of this project is to evaluate the processes related to the packaging of the current NASA food system being		

Task Description:	used for the International Space Station (ISS) and the Shuttle (STS). The evaluation will define the current processing parameters and identify ways to improve the processes. These parameters include those related to sealing, internal atmosphere of the package, and materials. Parallel to this evaluation, a literature review will be conducted in an effort to identify available materials with high potential for long duration missions, possible packaging defects and ways to prevent them in the future, ways to improve oxygen elimination, and the impact of these on product shelf life. Even though the current packaging system processing parameters have been documented, these parameters have not been evaluated from the perspective of improving packaging integrity and mass reduction. This is due largely to the success of the current packaging system using the current high payload vehicles such as the STS and Russian Progress vehicle. Payload for long duration missions of years rather than months, like ISS missions, will require a greater quantity of food in proportion to other supplies than do the missions of today. Thus, the need for reduced packaging mass becomes critical in order to execute future missions. The outline below describes specific processing points to be evaluated. These processing points are parts of the Total Packaging System. Any improvements that can be made in each of these areas may benefit the overall packaging system. ** Literature review o Identify high potential materials o Minimize pinholes and critical defects o Optimize Modified Atmosphere Packaging (MAP) flushing o Improve Shelf life ** Define existing packaging system o Food Processing -- Effects of Temperature on Materials o Heat seal -- Effects of temperature variation -- Seal Width (possible area of mass reduction) -- Quality 1. Defects 2. Seal Integrity o Vacuum -- Quality 1. Oxygen quantities -- MAP flushing quality 1. Using current system 2. Modified system 3. Gases other than nitrogen o Material quality -- Film -- Septum -- Gases (i.e. ppm 02) -- Packaging design/configuration ** Document possible improvements to existing system
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Optimal sealing parameters for each type of material currently being used by the space food system have already been incorporated into the packaging procedures. As new materials are added to the system, it is recommended that the procedure used in the heat seal evaluation be used to determine the optimal seal setting for each material. Although the seal reduction evaluation showed considerable mass saving, it is not recommend that the seals be reduced below the 3/8” standard. The safety and health of the crew is more important than minimal mass savings when it comes to a mission’s success. Reducing the seals would increase the potential for package failure from contaminants in the seal leading to food spoilage and food borne illness. The data that was gathered during the Modified Atmosphere Packaging (MAP) evaluation shows that the current methods used in the space food packaging system is inefficient and seals in a significant amount of air in some cases. It is highly recommended that the optimal equipment setting determined during the evaluation be incorporated as soon as possible to improve the quality of food in the space food system. The gusseted pouch design yielded very good results in making the rehydratable packages easier to produce while minimizing mass, volume, and waste. It reduced the production process from using three pieces of packaging equipment

Task Progress:	to one. It reduced the packaging from two pouches to one and decreased the total amount of packaging mass by approximately 63%. The literature review and MAP gas research stated that argon is a viable gas with great potential. Argon gas is more efficient, does not require a vacuum cycle, and can be used with liquid products, like the thermostabilized foods which currently have residual air in the headspace. More research and development is required for the use of argon gas for MAP applications because it is not widely used and requires equipment modifications. Also, it would need to be determined whether the levels of argon released from opened packages in an in enclosed environment would be hazardous to crew health. Several opportunities to improve the space food packaging system do exist. This proposal will gradually start incorporating the findings in each area as it is seen fit. These findings will be especially very crucial as missions increase in duration.
Bibliography Type:	Description: (Last Updated:)